Adidas US Sales Performance Analysis – Power BI Project

Project Overview

This project presents a comprehensive sales performance analysis of **Adidas US retail operations** using Power BI. The objective was to transform raw transactional data into meaningful business insights that support strategic decision-making, improve profitability tracking, and identify high-performing products, locations, and retail partners.

The dashboards were designed to replicate real-world business intelligence reporting standards, focusing on clarity, interactivity, and actionable insights.

Dashboard 1: Executive Sales Overview

Purpose

This dashboard provides leadership with a high-level snapshot of Adidas' overall business performance, helping stakeholders quickly assess revenue growth, profitability, and sales channel effectiveness.

Visual Explanations

Total Revenue Performance (KPI)
Shows the overall sales revenue generated across all regions and channels.
This KPI serves as the primary indicator of business scale and performance.

Total Operating Profit (KPI)
Highlights the net profit earned after operational costs.
It helps measure business efficiency and sustainability.

Total Units Sold (KPI)
Displays the total product volume sold.
This metric reflects product demand and market reach.

Average Selling Price per Unit (KPI)
Measures the average revenue earned per product sold.
Useful for evaluating pricing strategies and product value positioning.

Overall Profit Margin (KPI)
Shows the percentage of profit generated from total revenue.
It helps assess operational efficiency and cost management effectiveness.

Sales Performance Trend
Visualizes revenue growth across time (Year → Month).
This helps identify seasonal demand patterns and sales fluctuations.

Retailer Profit Contribution

Compares profit generated by each retail partner.
It highlights high-performing retail relationships and partnership opportunities.

Sales Distribution by Sales Channel

Evaluates performance across sales methods such as in-store, online, and outlet sales.
This helps identify the most effective customer engagement channel.

Revenue by Product Category

analysis sales contributions from product segments like footwear and apparel.
This helps guide inventory and product marketing strategies.

Geographical Sales Distribution

Displays state-wise revenue performance across the United States.
This supports regional expansion and targeted marketing decisions.

Dashboard 2: Product & Customer Insights

Purpose

This dashboard explores customer purchasing behaviour, product demand trends, and category-level performance to support merchandising and product planning decisions.

Visual Explanations

Total Category Revenue (KPI)
Displays cumulative revenue generated across all product categories.
Provides insight into product portfolio strength.

Total Product Volume Sold (KPI)
Measures the total quantity of products sold across categories.
Helps identify high-demand product segments.

Average Product Pricing (KPI)
Represents the average selling price across product lines.
Supports competitive pricing strategy evaluation.

Top Performing Product (KPI)
Identifies the individual product generating the highest revenue.
Supports focused marketing and inventory prioritization.

Top Performing Category (KPI)
Highlights the product category contributing the most sales revenue.
Helps determine where customer demand is strongest.

Category Demand by Gender
analyses product demand across gender segments within each category.
This provides insights into customer preference trends.

Product Revenue Performance
Compares revenue generated by individual product lines.
Supports performance benchmarking between products.

Customer Gender Sales Distribution
Shows the purchasing volume difference between male and female customers.
Helps identify demographic-driven marketing opportunities.

Category Revenue Share
Displays proportional revenue contribution by product categories.
Helps understand overall product portfolio balance.

Product Performance Summary
Provides detailed comparison of product-level sales, profit, and volume metrics.
Supports deep performance analysis and operational planning.

Dashboard 3: Regional & Retailer Profitability Analysis

Purpose

This dashboard focuses on geographical and partner-level profitability to help identify strong and underperforming markets and retail partners.

Visual Explanations

Total Regional Profit (KPI)

Shows cumulative profit generated across geographic regions.
Helps assess regional business strength.

Year-To-Date Revenue (KPI)
Displays cumulative revenue generated within the current year.
Supports performance monitoring against business targets.

Year-To-Date Profit (KPI)
Tracks profit accumulated during the current year.
Provides insight into profitability trends over time.

Most Profitable City (KPI)
Identifies the city generating the highest profit.
Helps determine successful market locations.

Top Profit Generating Retailer (KPI)
Highlights the retail partner contributing the most profit.
Supports partnership optimization strategies.

Regional Profit Comparison
Compares profitability across different regions.
Helps identify high-growth and underperforming territories.

Top 5 Cities by Profitability
Highlights the highest-performing cities based on profit.
Supports expansion and investment planning.

City-Level Profit Distribution
Provides a geographic visualization of profit performance across cities.
Helps detect regional profit concentration patterns.

Retailer Sales Performance
Compares sales contributions across retail partners.
Supports retail network performance evaluation.

Regional Sales and Profit Matrix
Provides detailed performance comparison across regions and states.
Helps identify localized business strengths and improvement opportunities.

Key Analytical Techniques Used

- Data Cleaning & Transformation using Power Query
- Time Intelligence Analysis (YTD, Trend Analysis)
- Dynamic Ranking using DAX
- KPI Performance Monitoring
- Geographic Data Visualization
- Interactive Filtering and Drill-Down Analysis

Key Challenges Faced

Data Standardization
Raw dataset contained mixed formatting and required transformation for consistent analysis.

Time Intelligence Implementation
Creating dynamic time-based analysis required building calculated date attributes and optimized measures.

Avoiding DAX Context Errors
Managing filter context and aggregation logic was critical to ensure accurate KPI calculations.

Performance Optimization
Balancing dashboard interactivity while maintaining performance required efficient measure design.

Business Storytelling
Transforming numerical data into meaningful business insights required careful visualization selection and layout planning.

Business Insights Derived

- Footwear categories consistently generated the highest revenue.
- Certain cities demonstrated significantly stronger profit performance, indicating regional market strength.
- Retail partner contribution varied widely, highlighting opportunities for partnership optimization.
- Sales channel performance indicated customer preference shifts toward specific buying methods.
- Seasonal trends revealed periods of high demand that can support inventory planning.

Tools & Technologies Used

- Power BI Desktop
- Power Query
- DAX (Data Analysis Expressions)
- Data Visualization & Dashboard Design
- Business Intelligence & Storytelling

Project Summary

This Power BI project successfully transforms Adidas sales data into an interactive and professional business intelligence solution. The dashboards provide stakeholders with a clear understanding of sales performance, customer behaviour, product demand, and geographic profitability.

By combining advanced data modelling, DAX calculations, and intuitive visualization design, this project demonstrates real-world data analytics skills focused on delivering actionable business insights.

All-in-One Code Snippet Document

SECTION 1 : All Power Query (M) columns
SECTION 2 : All DAX Calculated Columns (if used)
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Adidas US Sales Power BI – Complete Code Documentation

SECTION 1: Power Query (M Language) – Data Transformation Columns

These columns are created in **Power Query → Add Column → Custom Column**

Product Category Column

```
if Text.Contains([Product], "Footwear") then "Footwear"

else if Text.Contains([Product], "Apparel") then "Apparel"

else "Other"
```

Gender Column

```
if Text.StartsWith([Product], "Men") then "Men"

else if Text.StartsWith([Product], "Women") then "Women"

else "Unisex"
```

Year Column

```
Date.Year([Invoice Date])
```

Month Number Column

```
Date.Month([Invoice Date])
```

Month Name Column

```
Date.MonthName([Invoice Date])
```

Quarter Column

"Q" & Number.ToText(Date.QuarterOfYear([Invoice Date]))

SECTION 2: DAX Measures – Core Business KPIs

Create using:
Modeling → New Measure

Total Sales Amount

Total Sales Amount =

SUM('Data Sales Adidas'[Total Sales])

Total Profit Amount

Total Profit Amount =

SUM('Data Sales Adidas'[Operating Profit])

Total Units Sold

Total Units Sold =

SUM('Data Sales Adidas'[Units Sold])

Average Price per Unit

Average Price per Unit =

AVERAGE('Data Sales Adidas'[Price per Unit])

Average Operating Margin

Average Operating Margin =

AVERAGE('Data Sales Adidas'[Operating Margin])

Profit Margin Percentage

Profit Margin % =

DIVIDE([Total Profit Amount], [Total Sales Amount])

SECTION 3: Time Intelligence Measures

Sales YTD

```
Sales YTD =
TOTALYTD(
   [Total Sales Amount],
   'Data Sales Adidas'[Invoice Date]
)
```

Profit YTD

```
Profit YTD =
TOTALYTD(
   [Total Profit Amount],
   'Data Sales Adidas'[Invoice Date]
)
```

SECTION 4: Top Performance Measures

Top Product by Sales

```
Top Product (by Sales) =
VAR TopProductTable =
   TOPN(
      1,
      VALUES('Data Sales Adidas'[Product]),
      [Total Sales Amount],
      DESC
   )
RETURN
   CONCATENATEX(
      TopProductTable,
```

```
        'Data Sales Adidas'[Product],
        ", "
    )
```

Top Category by Sales

```
Top Category (by Sales) =
VAR TopCategoryTable =
    TOPN(
        1,
        VALUES('Data Sales Adidas'[Product Category]),
        [Total Sales Amount],
        DESC
    )
RETURN
    CONCATENATEX(
        TopCategoryTable,
        'Data Sales Adidas'[Product Category],
        ", "
    )
```

Top City by Profit

```
Top City by Profit =
VAR TopCityTable =
    TOPN(
        1,
        VALUES('Data Sales Adidas'[City]),
        [Total Profit Amount],
        DESC
    )
RETURN
    CONCATENATEX(
```

```
    TopCityTable,
    'Data Sales Adidas'[City],
    ", "
  )
```

Top Retailer by Profit

```
Top Retailer by Profit =
VAR TopRetailerTable =
  TOPN(
    1,
    VALUES('Data Sales Adidas'[Retailer]),
    [Total Profit Amount],
    DESC
  )
RETURN
  CONCATENATEX(
    TopRetailerTable,
    'Data Sales Adidas'[Retailer],
    ", "
  )
```

SECTION 5: Ranking Measure

City Profit Rank

```
City Profit Rank =
RANKX(
  ALL('Data Sales Adidas'[City]),
  [Total Profit Amount],
  ,
  DESC
```

)

SECTION 6: Location Based Measures

Sales by Location

Sales by Location =

[Total Sales Amount]

Profit by Location

Profit by Location =

[Total Profit Amount]

SECTION 7: Advanced Label Measure (Optional)

Top City Profit Label

```
Top City by Profit (Label) =
VAR T =
   TOPN(
      1,
      SUMMARIZE(
         'Data Sales Adidas',
         'Data Sales Adidas'[City],
         "Profit", [Total Profit Amount]
      ),
      [Profit],
      DESC
   )
RETURN
   CONCATENATEX(
      T,
      'Data Sales Adidas'[City] & " - " &
```

```
    FORMAT([Profit], "$#,##0"),
    ", "
)
```

SECTION 8: Model Sorting Requirement

After loading data:

👉 Sort Month Name by Month Number
This ensures chronological order in visuals.

SECTION 9: Data Model Assumptions

- Invoice Date used for Time Intelligence
- Product column used to derive:
  - Category
  - Gender
- Sales and Profit calculations based on aggregation measures

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SECTION 10: Best Practice Notes

✓ Measures used instead of calculated columns for aggregations
✓ Naming conventions maintained to avoid ambiguity
✓ Time intelligence implemented using YTD logic
✓ Dynamic ranking created using TOPN and RANKX
✓ Data cleaning performed in Power Query

End of Documentation

This document contains the complete transformation and analytical logic used in the Adidas US Sales Power BI Project.

======================== Thank You ========================